

Stage 2

To start with the execution of Stage 2 you must have done Stage 1, for this Stage you will use ngrok which is a service that allows you to create the local server in a subdomain and then be able to view it outside the LAN through the internet.

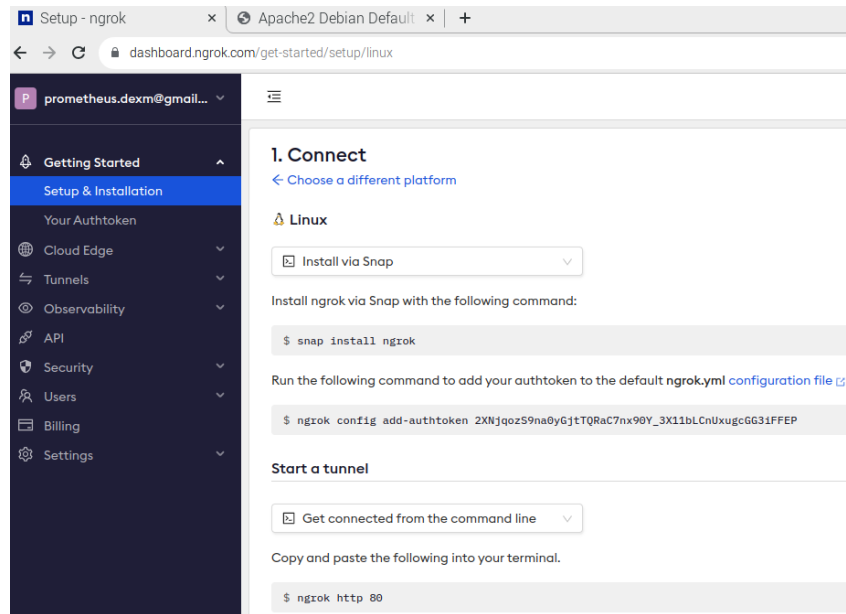


Image 1. Ngrok.

In the browser of the Promedex device you will find in favorites the ngrok page to enter the administration console and see relevant information of the connections, in the image you can see the installation process, however it is the last command line that will be used to start with the service, It should be noted that the scenarios are automated, so to start the NGROK service, reserved commands will be used that point directly to the preconfigured local resources.

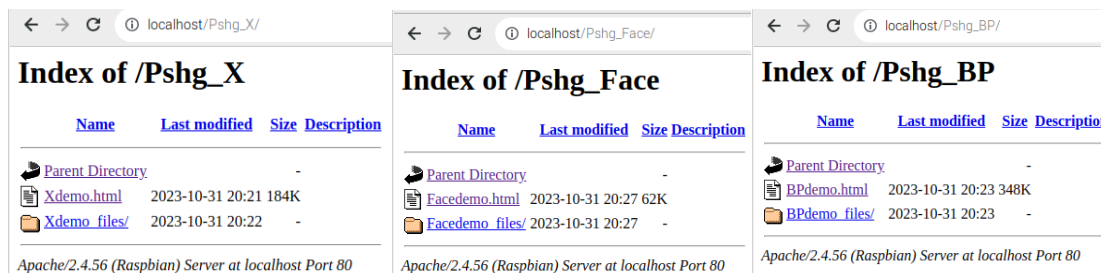


Image 2. Local resources.

The Promedex device has 3 web pages loaded, one from the social network Facebook, another from the social network X and the last one from the banking service of Banco de Pichincha, in the case of the latter it is necessary to remember the command line that allows you to enter this resource that is

[localhost/Pshg_BP](#) this is because this resource is not linked to the ngrok service to avoid problems of any kind.

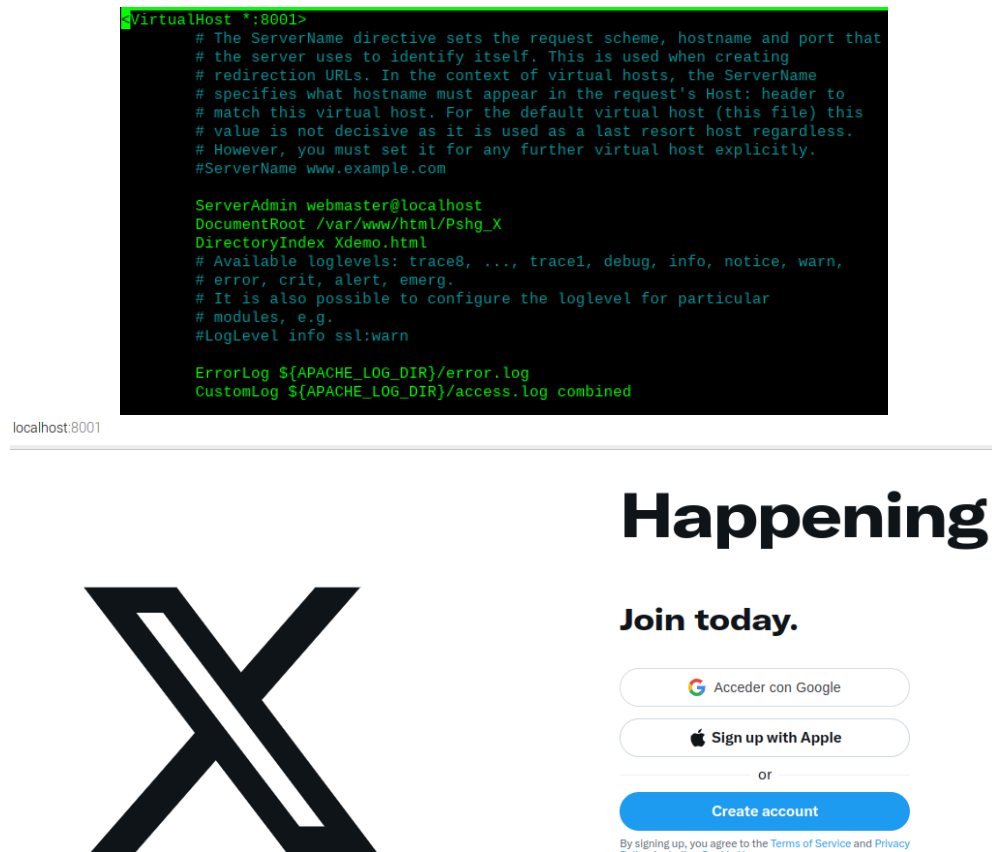


Image 3. Port 8001.

The social network X page is associated with port 8001, in the image you can see the entry from the localhost specifying port 8001 using [localhost:8001](#).

localhost:8000

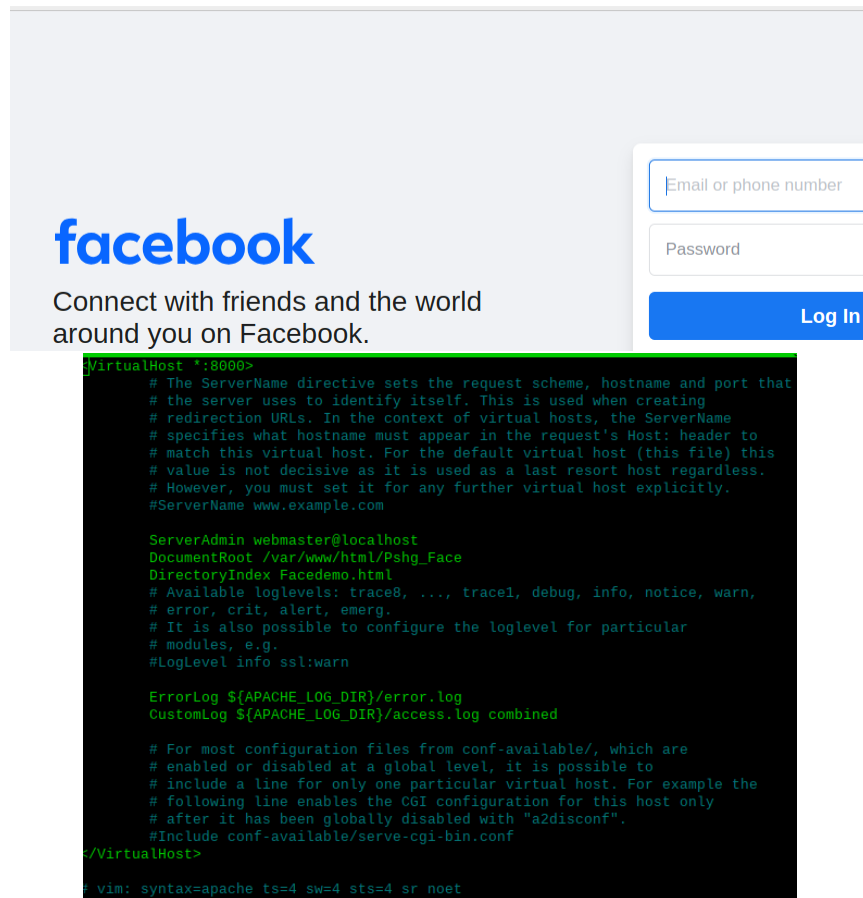


Image 4. Port 8000.

To start the Facebook social network page, you must specify port 8000 that is linked to that resource with the `localhost:8000` command.

To start the demo to the participants you must choose which page will be displayed and the appropriate reserved command will be used to start the ngrok service.

In the case that you want to start the page of social network X, you must use the reserved command `x` or `X` and it will automatically start the ngrok service, you must wait for the service to generate a public link, which is the one found in the [forwarding field](#).

```
ngrok (Ctrl+C to quit)

Introducing Pay-as-you-go pricing: https://ngrok.com/r/payg

Session Status      online
Account             prometheus.dexm@gmail.com (Plan: Free)
Version             3.4.0
Region              South America (sa)
Latency              138ms
Web Interface        http://127.0.0.1:4040
Forwarding           https://f737-157-100-140-163.ngrok-free.app -> http://localhost:8001

Connections          ttl    opn    rt1    rt5    p50    p90
                    21     0      0.05   0.05   0.82   5.08

HTTP Requests
-----
GET /sw.js                                                    404 Not Found
GET /Xdemo_files/appleid.auth.js                             200 OK
GET /Xdemo_files/client                                       200 OK
GET /Xdemo_files/main.595d9dda.js                             200 OK
GET /Xdemo_files/en.6e6b21ca.js                               200 OK
GET /Xdemo_files/style                                         200 OK
GET /Xdemo_files/analytics.js                                  200 OK
GET /Xdemo_files/vendor.e761876a.js                           200 OK
GET /Xdemo_files/button.html                                   200 OK
GET /Xdemo_files/shared-loader.Typeahead-bundle.Communities-bundle.Delegate-bundle.Ocf-bundle.LoggedOutHome-bundle.Search-bund.23794d0a.js 200 OK
```

Image 5. Start ngrok.

Once the ngrok service is started, the public link must be copied into the browser's search bar, it will redirect us to a page as shown in Image 6. In which participants could be explained not to enter personal information, as the ngrok service warns.

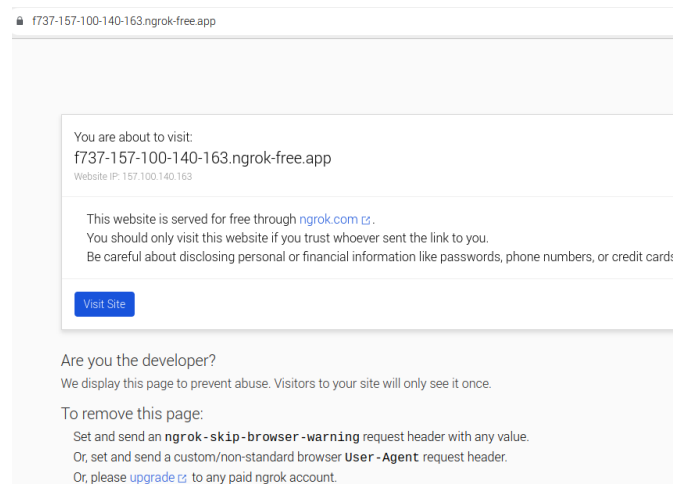


Image 6. Ngrok security.

You must access the page by clicking on [visit site](#), so you will be able to access the selected page.

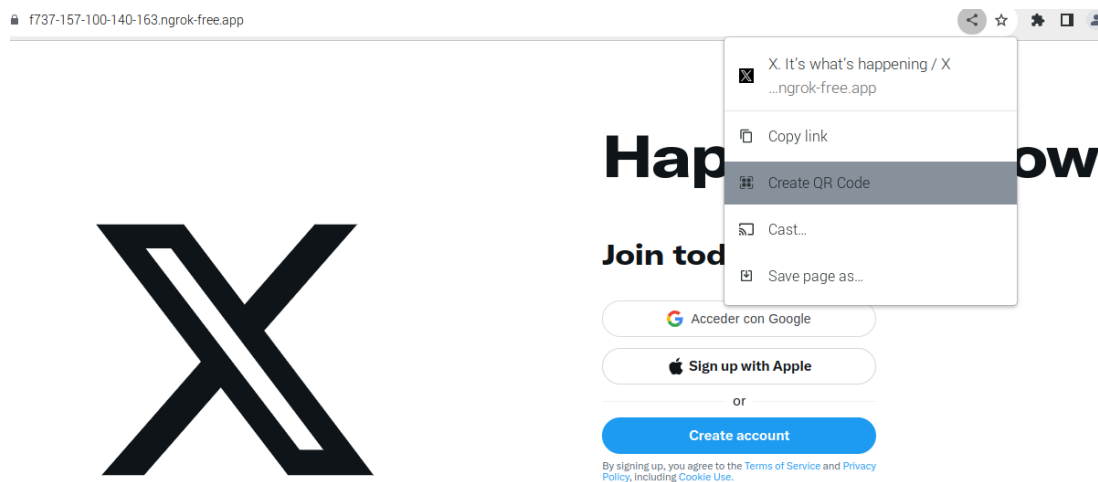


Image 7. Page X.

Once inside the page we can notice that the link of the page contains an identifier of ngrok, to allow participants to interact with the page a QR code must be generated from the share button in the search bar.

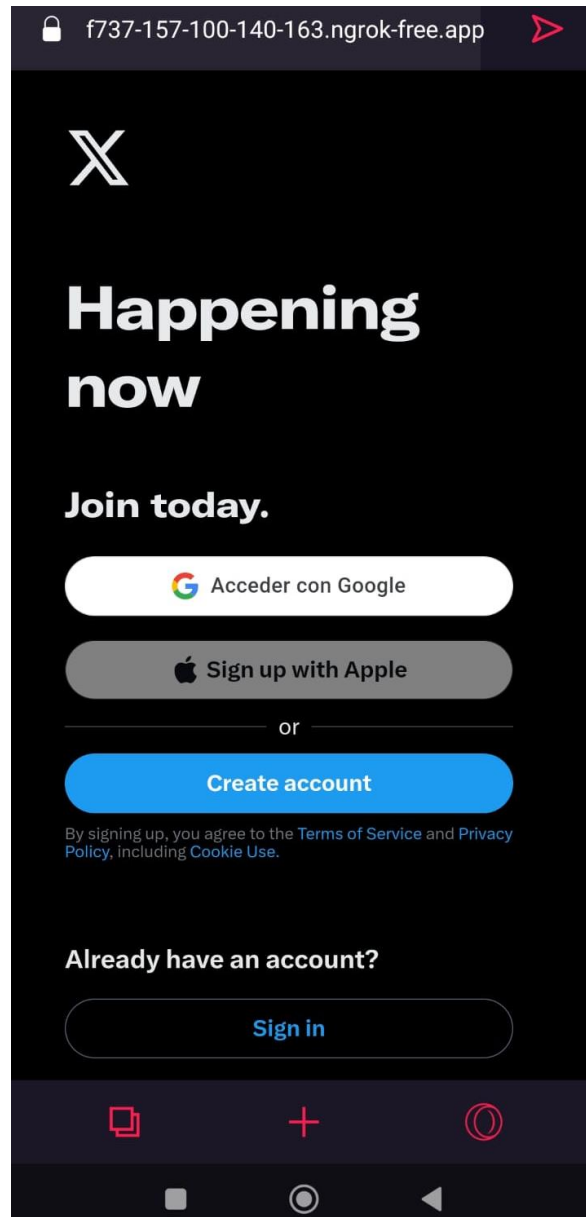


Image 8. Page X.

The page can be accessed with the shared link as seen in image 8, which was accessed from a smartphone.

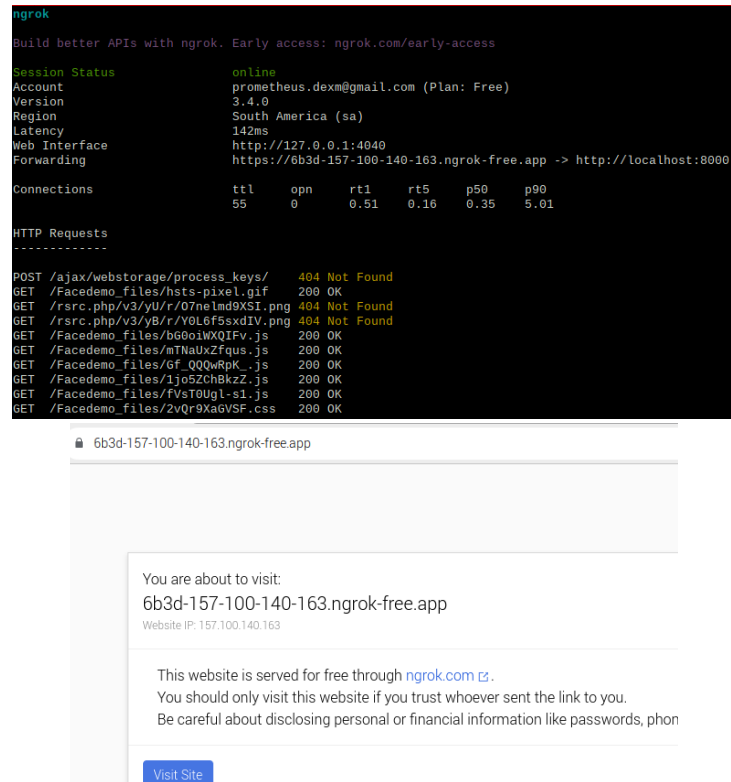


Image 9. Ngrok Facebook.

You can also start the Facebook page, using the reserved **command** `face` or `Face` this will automatically activate the ngrok service, you must wait until the public link is available and then copy it to the browser, the ngrok security message will be displayed and the content will be accessible.

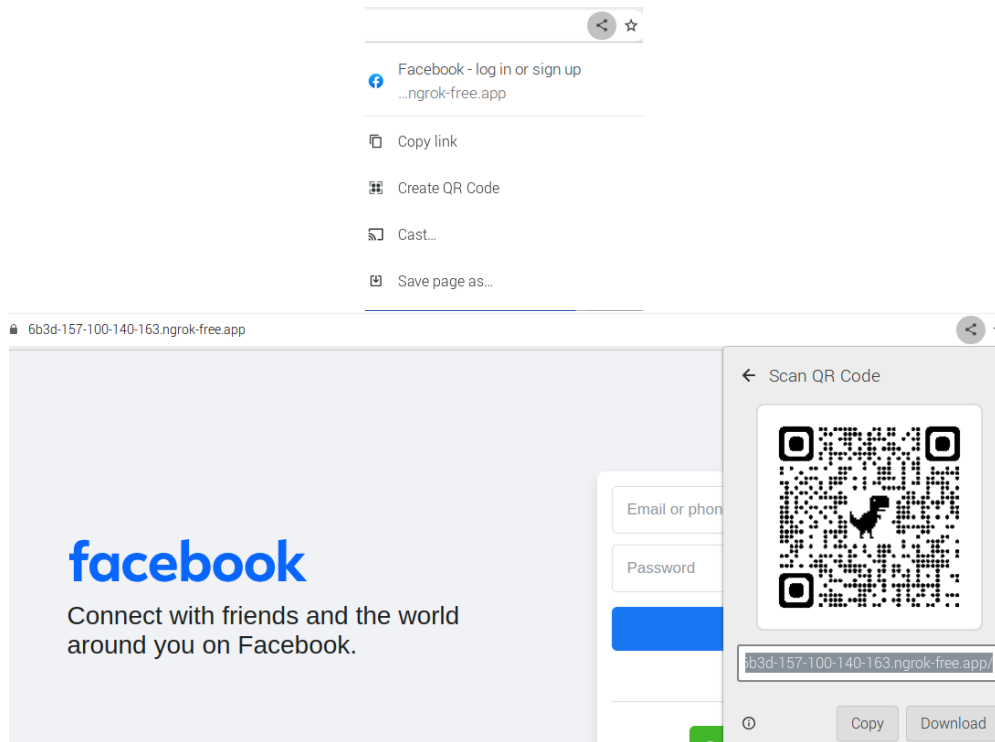


Image 10. Share QR.

In the address bar you will find the share icon, you can choose to create a QR code, by selecting it, as you can see in **image 10**. a QR code is displayed that can be scanned by participants to access the page.

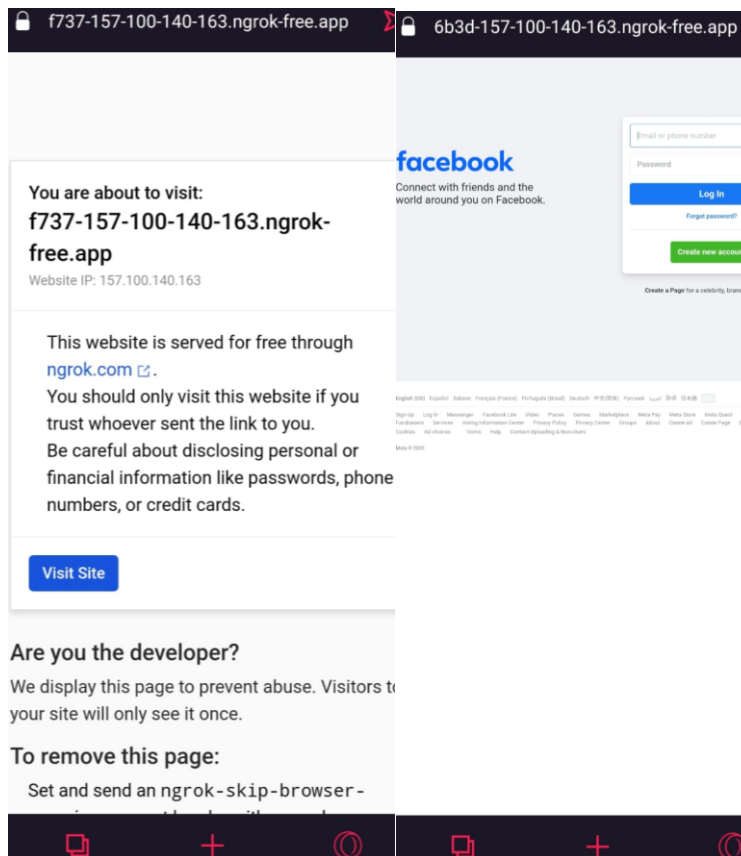


Image 11. QR access.

The image shows access to the page from the QR code from an Android device.

```
spn@raspberrypi:~$ S con
Address      HWtype  HWaddress      Flags Mask    Iface
157.148.63.205      (incomplete)          wlan0
129.226.3.47       (incomplete)          wlan0
183.47.125.78      (incomplete)          wlan0
101.32.104.4       (incomplete)          wlan0
123.151.56.102     (incomplete)          wlan0
101.32.104.41     (incomplete)          wlan0
101.32.118.25     (incomplete)          wlan0
Xiaomi-11-Lite-5G-NE ether 5a:de:f7:63:2f:96 C    wlan0
120.241.190.60     (incomplete)          wlan0
121.51.162.69     (incomplete)          wlan0
^C
spn@raspberrypi:~$ S fcon
? (157.148.63.205) at <incomplete> on wlan0
? (129.226.3.47) at <incomplete> on wlan0
? (183.47.125.78) at <incomplete> on wlan0
? (101.32.104.4) at <incomplete> on wlan0
? (123.151.56.102) at <incomplete> on wlan0
? (101.32.104.41) at <incomplete> on wlan0
? (101.32.118.25) at <incomplete> on wlan0
Xiaomi-11-Lite-5G-NE (10.0.1.19) at 5a:de:f7:63:2f:96 [ether] on wlan0
? (120.241.190.60) at <incomplete> on wlan0
? (121.51.162.69) at <incomplete> on wlan0
dns.google (8.8.8.8) at <incomplete> on wlan0
dns.google (8.8.4.4) at <incomplete> on wlan0
? (43.154.240.84) at <incomplete> on wlan0
? (175.27.38.131) at <incomplete> on wlan0
? (109.244.172.79) at <incomplete> on wlan0
raspberrypi.mshome.net (192.168.137.232) at <incomplete> on wlan0
? (416.130.223.103) at <incomplete> on wlan0
LAPTOP-17B3FRV9.mshome.net (192.168.137.1) at <incomplete> on wlan0
? (43.130.30.240) at <incomplete> on wlan0
```

Image 12. User verification.

You can check the connected users with the `con` and `fcon` command to see the connections generated when the participants are connected to the network of the promedex device.



Imagen 13. Local host BP.

To end the demonstration, the localhost must be accessed so that participants can observe the mechanisms used by attackers to make a user believe that they are entering their passwords and carrying out criminal activities.

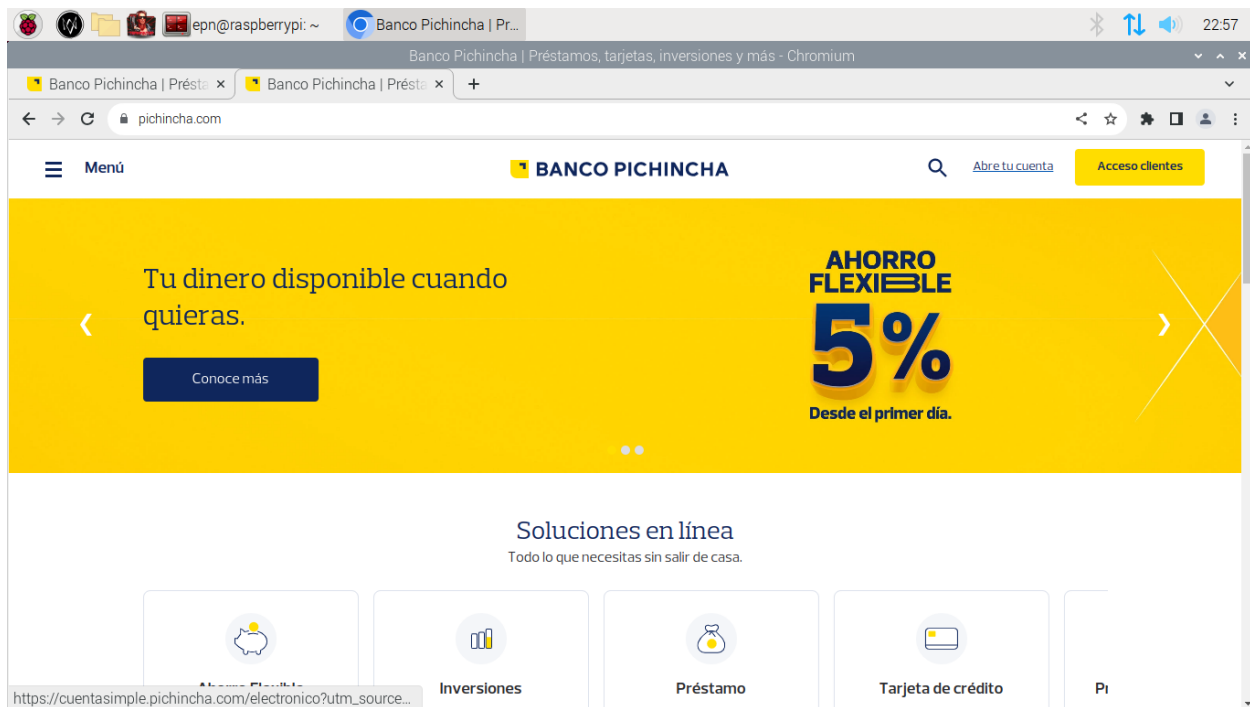


Image 14. Real page.

The exhibitor must make comparisons on the real and cloned pages so that participants can see the obvious differences on the pages such as the certificates and from the same entry link.

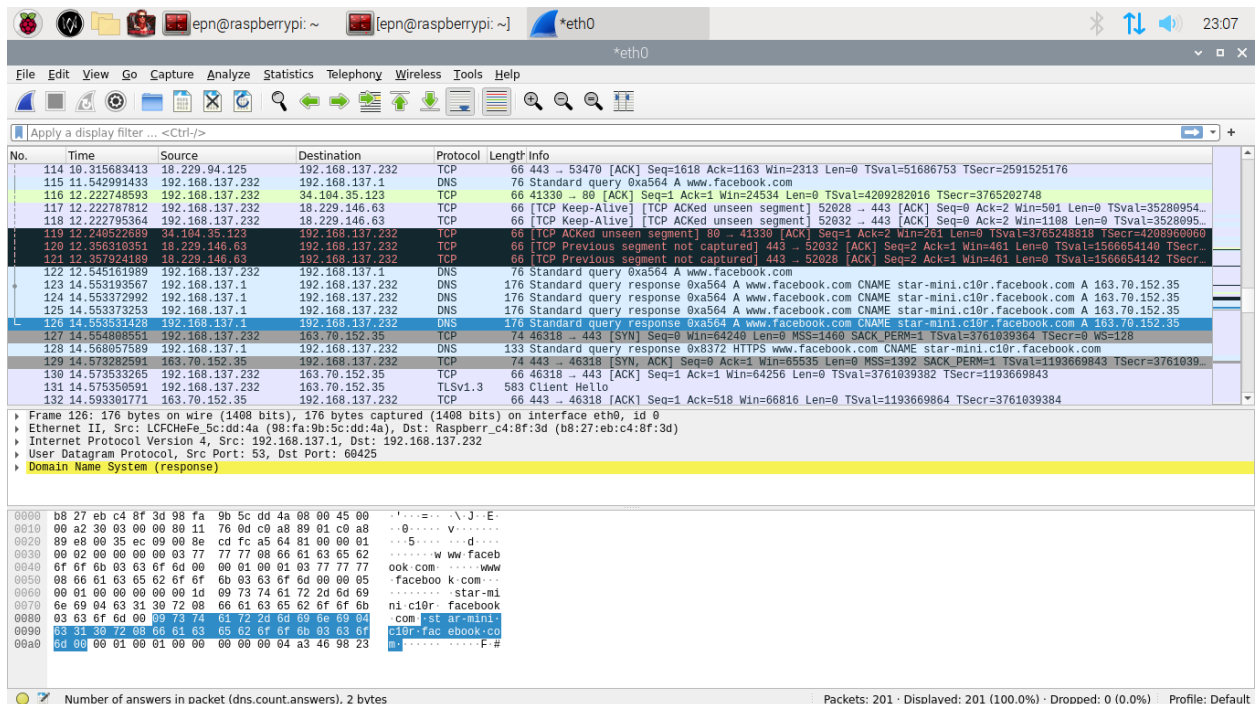


Image 15. Traffic capture.

Traffic captures can be made in the interaction with all the aforementioned pages to explain to participants the methodology used by criminals to steal personal information or do social engineering so that they can enter their personal data without making the victim suspect that their credentials are being stolen.

The offline version of this Stage can be executed by entering the local resources mentioned above, these are hosted on the Web server of the Promedex device.

Prepared by JS.